HEAT SENSOR CIRCUIT

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Abstract: This project presents a temperature-sensing circuit using a thermistor and operational amplifier comparator, which activates a buzzer and LED through a transistor when a preset temperature threshold is exceeded

I. Introduction

The purpose of this project is to develop a basic heat sensor circuit using a thermistor, an operational amplifier (Op-Amp), and a transistor. This circuit detects temperature variations and triggers an alert system (LED and buzzer) when the sensed temperature exceeds a predetermined threshold. Such systems are commonly used in fire alarms, temperature-controlled devices, and industrial safety systems.

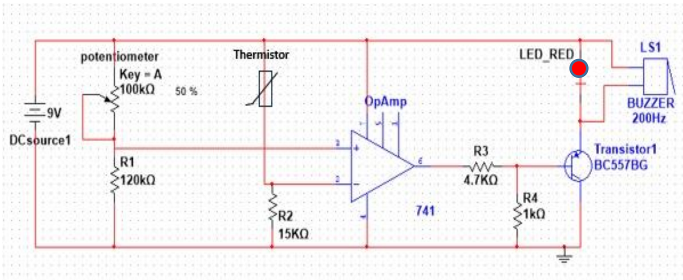
II. Literature Review

Temperature sensors, particularly thermistors, have been widely utilized in electronics for thermal monitoring. Thermistors are resistors whose resistance decreases with increasing temperature (NTC type). Amplifiers like the LM741 operational amplifier are used to compare voltage levels corresponding to temperature variations. When temperature rises above the set limit, the comparator circuit activates actuators such as LEDs or buzzers. The integration of a transistor switch further allows for controlling external components based on the Op-Amp output.

III. Components Used in This Circuit

- **DC Power Supply (9V):** Provides power to the entire circuit.
- **Potentiometer (1 x 100kΩ):** Used to set the reference voltage and sensitivity.
- **Thermistor:** Senses the temperature change.
- **Resistors:** Current limiting and voltage division.
    - R1= 120K ohm
    - R2= 15K ohm
    - R3= 4.7K ohm
    - R4= 1K ohm
- **LM741 Operational Amplifier:** Functions as a comparator.
- **Transistor (BC557 - PNP):** Acts as a switch to control the buzzer and LED.
- **LED (Red):** Visual indication of high temperature.
- **Buzzer (200 Hz):** Audible alert for high temperature condition.

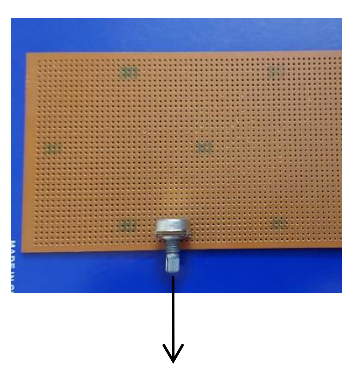
IV. Circuit Diagram:



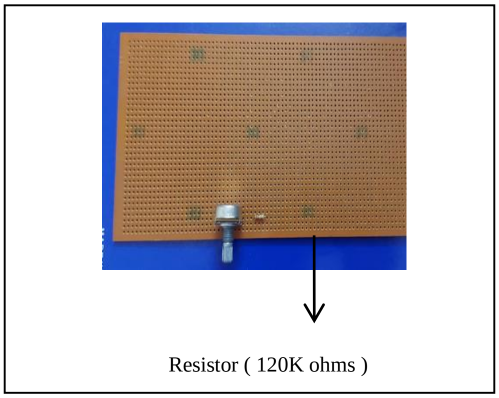
V. Methodology (Step-by-Step):

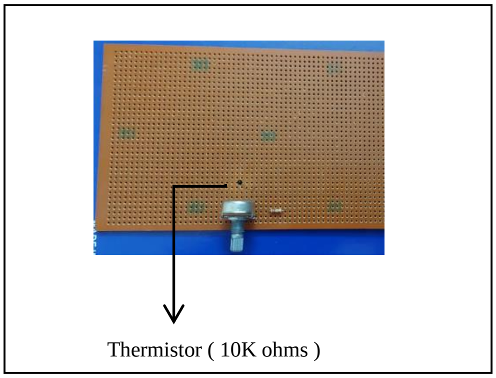
- **Power Supply Setup:**
  - A 9V DC power source is connected to power the entire circuit.
  - Ensure proper connection of ground (GND) and Vcc for all components, especially the Op-Amp.
- **Reference Voltage Configuration:**
  - A **potentiometer** is connected and adjusted to set a fixed reference voltage.
  - This reference voltage is fed to the **non-inverting (+) input** of the **LM741 Op-Amp**.
- **Sensor Voltage Divider Network:**
  - A **120kΩ resistor** is connected in series with the **thermistor** (NTC type).
  - This combination forms a voltage divider to generate a variable voltage depending on ambient temperature.
  - The junction between the 120kΩ resistor and the thermistor is connected to the **inverting (–) input** of the Op-Amp.
- **Comparator Stage (Op-Amp LM741):**
  - The **LM741** compares the voltage from the thermistor divider (inverting input) and the potentiometer (non-inverting input).
  - When the voltage from the thermistor side drops **below** the reference voltage (due to increased temperature), the Op-Amp output switches HIGH.
- **Output Drive Stage:**
  - The output of the Op-Amp is connected to the base of a **PNP transistor (BC557)** via a **4.7kΩ resistor** (green arrow).
  - The **collector** of the transistor connects to both an **LED** and a **buzzer**, providing visual and audio signals.
- **Load Resistor:**
  - A **15kΩ resistor** (red arrow) is placed appropriately to limit current, typically either in series with the LED/buzzer or across the base-emitter junction to control biasing current.
- **Trigger Response:**
  - When the output of the Op-Amp is HIGH (due to high temperature), the PNP transistor is **turned ON**, allowing current to flow through the buzzer and LED, thereby activating them.
  - When the temperature is normal, the Op-Amp output remains LOW, keeping the transistor OFF.

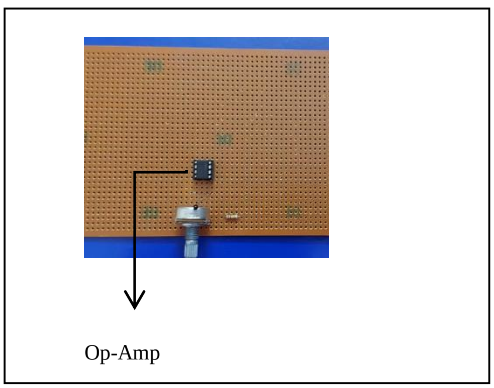
V. Pictures during experiment:

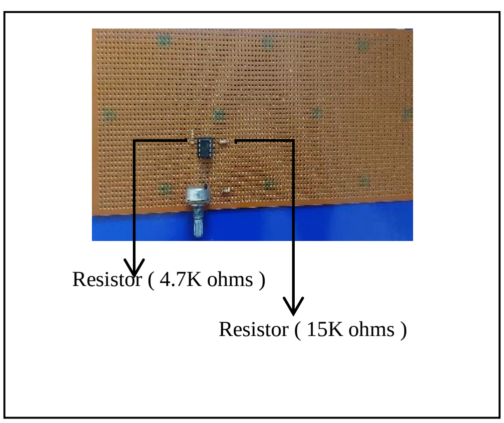


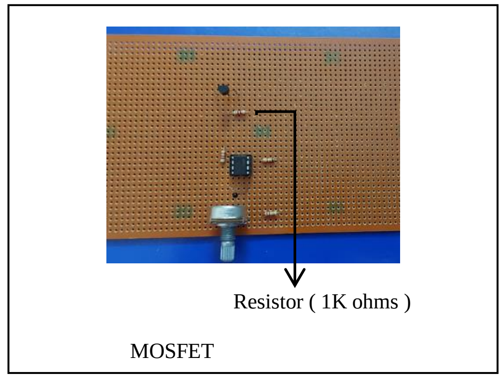
Potentiometer

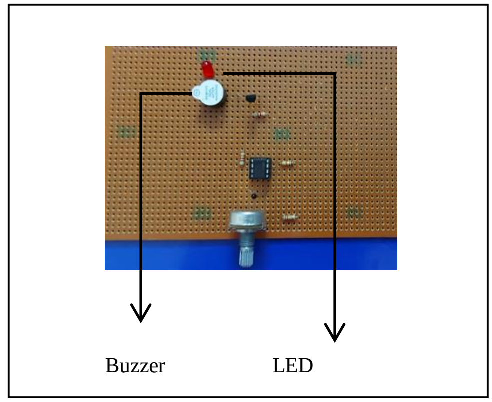


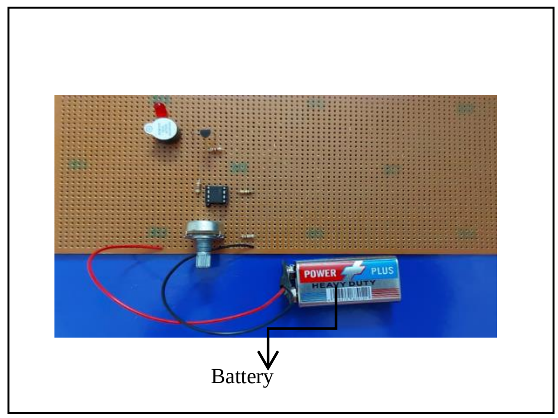












VI. Design Explanation

- **Voltage Divider with Thermistor:** Adjusts voltage at the Op-Amp's inverting input based on ambient temperature.
- **Potentiometer Reference:** Sets a threshold voltage at the non-inverting input of the Op-Amp.
- **Op-Amp Comparator (LM741):** Compares the two voltages; outputs HIGH when thermistor voltage is lower than reference.
- **Transistor Switch (BC557):** Activates the buzzer and LED when the Op-Amp output is HIGH.

- **Output Alert System:** Comprises a buzzer and an LED, both of which get activated upon high temperature detection.

VII. Result

Upon simulation, the circuit successfully triggers the LED and buzzer when the temperature exceeds a predefined level. This demonstrates that the comparator circuit is functioning as expected, and the transistor is effectively acting as a switch.

VIII. Application

- Fire detection systems
- Industrial equipment temperature monitoring
- Battery overheat protection
- Server room temperature alert
- Electronic device thermal management

IX. Advantages

- Simple and cost-effective design
- Adjustable temperature threshold
- Clear visual and audible alert system
- Can be integrated into other systems
- Low power consumption

X. Disadvantages

- Limited to basic threshold detection (no precise temperature reading)
- Sensitivity may vary with thermistor tolerance
- Not suitable for very high-temperature environments without modification
- Uses a general-purpose Op-Amp (LM741), which is not optimized for fast switching.

X1. Conclusion

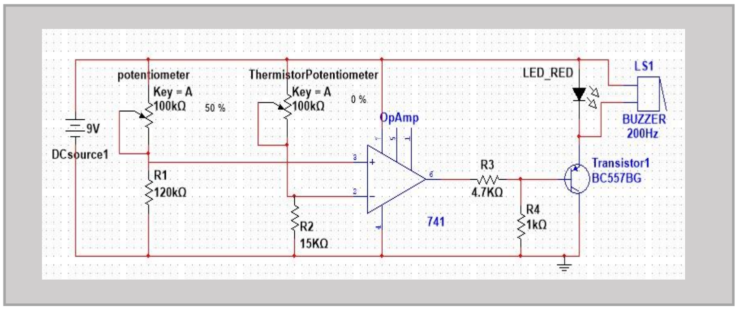
The heat sensor circuit developed in this project successfully demonstrates the use of a thermistor and comparator (LM741) to detect and respond to changes in temperature. By adjusting the potentiometer, the user can set a threshold temperature. Once the ambient temperature rises above this level, the circuit activates a buzzer and LED to provide immediate alerts. This makes the system suitable for basic thermal monitoring applications. The project illustrates core principles of analog electronics, including voltage dividers, Op-Amp comparator operation, and transistor switching.

XI. Circuit Simulation

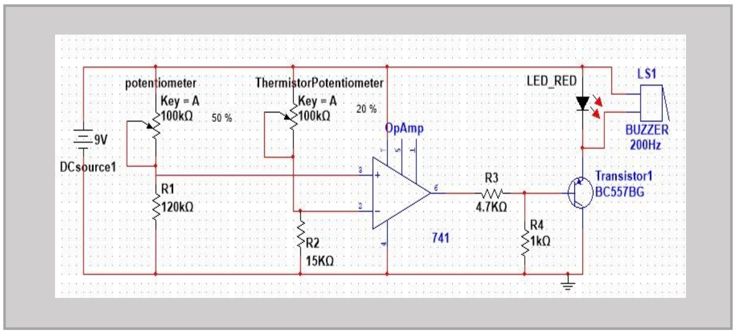
The circuit was simulated using Proteus. In the simulation:

- A **9V DC source** powers the circuit.
- The **thermistor's resistance** changes with simulated temperature input.
- When the temperature increases (simulated by reducing the thermistor resistance), the voltage at the inverting input of the LM741 falls.
- Once this voltage drops below the reference voltage set by the potentiometer, the Op-Amp output becomes HIGH.
- This HIGH output turns on the **BC557 PNP transistor**, allowing current to flow through the **LED and buzzer**, indicating a high-temperature condition.

The simulation confirms that the circuit responds correctly to heat by activating visual and audio alerts. It validates the design's functionality and effectiveness in temperature detection scenarios.



✧ **CIRCUIT SIMULATION BEFORE DETECTING HIGH TEMPERATURE**



✧ **CIRCUIT SIMULATION AFTER DETECTING HIGH TEMPERATURE**